

REPORT ON STORY GENERATOR (SMART AGENT MODULE)

Prepared By:

Satvik Sahu
2328118
Kalinga Institute of Industrial Technology
Branch: Computer Science and System Engineering
Subject: Minor Project

Module: Story Generator (Agent 6)
Project: Smart Athlete Support System

1. Introduction

In the Smart Athlete Support System, multiple intelligent agents work together to assist differently-abled athletes. The Story Generator (Agent 6) is responsible for converting structured athlete data into a meaningful and impactful narrative.

This module takes processed data from previous agents and transforms it into a well-written story suitable for NGOs, sponsors, and stakeholders. The goal is to present athlete journeys in a compelling and human-centric way.

2. Objective of the Module

The main objectives of the Story Generator are:

- To convert structured athlete data into a readable story
 - To highlight athlete struggles, achievements, and goals
 - To create emotionally engaging and professional narratives
 - To support multilingual output (English, Hindi, Odia)
 - To work even when some input data is missing
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3. Working of Story Generator

The Story Generator operates in three main stages:

Step 1: Input Collection

The module receives structured input from previous agents, including:

- Name and Age
- Location
- Disability details
- Sport and achievements
- Coach and training details

- Goals and aspirations
 - Financial or support needs
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Step 2: Smart Processing

The system uses rule-based logic to:

- Extract relevant fields from input
- Handle missing information intelligently
- Adjust sentence structure dynamically
- Apply tone selection (Professional / First-person)

It ensures that only available data is used, avoiding incorrect or incomplete sentences.

Step 3: Story Generation

The module generates:

- A 200-word narrative story
 - A structured athlete profile
 - Context-aware paragraphs covering:
 - Background and challenges
 - Training and performance
 - Achievements and goals
 - Support needs and future potential
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4. Key Features

- Dynamic story generation from structured data
 - Handles incomplete or missing input gracefully
 - Tone-based storytelling (Professional / First-person)
 - Multi-language support:
 - English
 - Hindi
 - Odia
 - Structured athlete profile generation
 - Clean and readable narrative output
 - Fully offline functionality (no external AI API required)
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5. Technologies Used

- HTML
 - CSS
 - JavaScript
 - Node.js (Backend API)
 - Express.js (REST API framework)
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6. Input Methods

The module supports:

- Text-based athlete descriptions
 - Structured data from previous agents
 - Combined inputs
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7. Output Generated

The Story Generator produces:

1. Athlete Profile (Structured Data)

- Name
- Age
- Location
- Disability
- Sport
- Coach
- Achievements
- Goals
- Support Needs

2. Narrative Story

- 180–220 word story
 - NGO-ready format
 - Human-readable and impactful
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8. Challenges Faced

- Generating natural language without AI APIs
 - Handling missing or incomplete data
 - Maintaining story quality and consistency
 - Supporting multiple languages
 - Avoiding repetitive or robotic sentences
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9. Improvements Made

- Enhanced sentence construction logic
 - Added dynamic paragraph generation
 - Integrated language translation feature
 - Improved readability and structure
 - Added structured profile output
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10. Importance of the Module

The Story Generator is crucial because:

- It converts raw data into meaningful stories
 - Helps NGOs understand athlete needs quickly
 - Improves communication and funding chances
 - Acts as the final presentation layer of the system
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11. Future Scope

- Integration with AI-based language models
 - Improved translation accuracy
 - Voice narration feature
 - PDF export functionality
 - Advanced storytelling personalization
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12. Conclusion

The Story Generator successfully transforms structured athlete data into compelling narratives. It enhances the overall system by making the information more understandable, engaging, and useful for decision-making.

13. Personal Learning

From this module, I learned:

- How to design rule-based content generation systems
- Basics of REST API development
- Handling dynamic data in real-time applications
- Importance of user experience and readability
- Building practical solutions for real-world problems